Working on LongEval: Multilingual Retrieval, ReJudgement, and Language Effects

- Course / module: Web Information Retrieval
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Background

During the judging of our documents, we noticed that each of us received a surprisingly large number of non-english documents. Some of these already had partial english translations. In some cases, the translation was immediately visible at the beginning of the document, while in the others it was hidden further down in the text without any clear distinction between the original language and the english part. This raised the question of whether we, or any of the other judges missed potentially relevant documents due to our limited language knowledge. Although the queries were written in english, scientific literature is often multilingual. And Since automatic translation has become much more common language is less of a barrier for relevance. Based on that we questioned whether relevant documents might have been judged as irrelevant simply because their original title or abstract was not written in english.

Based on that idea we formed some research questions and hypotheses which we could answer with the available data:

Research Questions:

Does translating non-English documents that were originally judged as irrelevant into English change their relevance judgements, and do these re-judgements differ between human judges and an LLM-based judge?

Hypotheses:

H1: Translating previously irrelevant non-English documents into English increases the proportion of documents receiving a relevant judgement (≥ 1).

H2: Human and LLM-based relevance judgements differ significantly for translated non-English documents.

Methodology

Firstly, we filtered the documents that were both non-english and originally judged as irrelevant. We focused on these documents because we assumed that documents judged relevant were already understandable by the original judges.
From the initial filtering process, we identified 92 documents that matched these criteria – irrelevant and non-english. These documents were divided into three batches with each team member receiving one batch. For each document we performed the following steps:

1. Translate the Titles and Abstract using DeepL
2. Re-Judge each document according to the topics we were provided
3. Added additional information about the document, including the original language, whether an English translation already existed, if yes where is the translation at the top (and therefore clearly visible) or at the bottom of the text (and therefore more hidden)

After that step, we used the english translations created with DeepL to let an LLM judge the complete dataset using the MTO Principle[1]. For this we used the topig-gen GitHub[2] implementation that was already used for the LLM topic and qrel generation of LongEval. Sadly, we were not able to use the same LLM, as the code called for Gemini 2.5 Flash, which is no longer freely available since March 2026 and we were not willing to pay for its use. And since we wanted to use the same code, we instead switched to Mistral Medium.

In the final step, we compared the original Qrels with the new human judgements and the LLM judgements. We analyzed how many documents changed their relevance score, how human and LLM judgements differed and whether language related patterns could explain why documents were originally overlooked. We also focused on finding out if the baseline runs from LongEval would be evaluated differently with each Qrel.

Results

Human Relabelling Results

After filtering the LongEval-Sci corpus, we identified 92 non-english documents that had originally been judged as irrelevant. After translating and rejudging these documents, we found that 64 out of the 92 documents (69,6%) received a different relevance judgement. While 28 documents remained irrelevant, 41 documents were judged as relevant and 23 as highly relevant by us.

We also found out that the documents covered a wide range of languages with Spanish (27x) and Indonesian (23x) being the most common language. We also found that

[1] (Thomas, P., Spielman, S., Craswell, N., & Mitra, B., 2024)
[2] (Keller, 2026)

language alone did not explain why these documents were missed. Although 57 of the 92 documents (62%) already contained some English text, this translation was located at the bottom of the document in 35 cases (61%), making it easy to overlook.

We also took a closer look at the documents with english translations at the top. There were 22 such documents and all of them had originally been judged as irrelevant. After rejudging them, we considered 15 of them (68%) to be relevant. We also noticed that 16 of these 22 documents still had a non-english title. This suggests that the orignal judges may have relied mainly on the document title and therefore skipped documents that actually contained english information further down.

We also compared our judgements with LLM-based judgements. Overall, both judged a similar number of documents as irrelevant but there were clear differences in how relevant the remaining documents were considered. While our team mostly assigned the label relevant (1), the LLM much more often chose highly relevant (2). This shows that the biggest difference between our and the LLM judging was not deciding whether a document was relevant but rather how strongly relevant it should be considered.

Impact on Retrieval Evaluation

To see whether these new judgements would actually change the evaluation of retrieval systems, we repeated the LongEval evaluation using three qrels: the original qrels, our human qrels and the qrels generated by Mistral Medium.

We carried out the evaluation using Ranx, a module which we were more familiar with and that offers comparison between runs. We evaluated both the LongEval Baseline runs one of which was a BM25 run while the other a Dense run. We looked at the metrics MAP, Reciprocal Rank, nDCG, Recall and Precision at different cut-offs (from k@5 increasing to k@100).

For both runs evaluating them with all three baselines showed a difference between the rejudged qrels and the original qrels. Interestingly, the human and llm qrels showed very little difference.
While for the B25 run most metrics were almost unaffected and the recall even negatively affected by the new qrels, the Dense retrieval run showed more improvement. The nDGC@k was slightly improved for the BM25 before the 50 mark while for the Dense retrieval run it was improved by a constant degree. Suggesting that here one or multiple documents changed retrieval judgements which were retrieved early on.

Overall, it seems that more documents which were retrieved were judged relevant by both us and the llm. When investigating the small difference, it showed that the llm and us disagreed mostly on documents with very high ranks which most time fell outside the metrics we looked at. The few times the rank was low, the relevance difference was as low as well and this was visible in metrics such as nDCG which did show some variation between the qrels.

Conclusion

While our results were not as hoped, we firmly believe that non-English documents should be translated for the judges before the judgement phase. Assuming monolingual user is faulty in our modern world especially with the tools available for translation. Therefore, a judge should always consider all documents and never deem one irrelevant on language base alone.

Who	Did What
Muhammed	First Corpus filtering / Data preparation
Janina and Maryam	First Report
Everyone	Relabeling 1/3 of the corpus
Muhammed & Maryam	Analysis LLM vs. Human Qrels
Janina	Run Evaluation
Maryam	Presentation